I. What is R and what are its primary uses?

II. What are the main data types in R?

III. What is the difference between a vector and a list in R?

IV. What is a data frame in R, and how is it different from a matrix?

V. How do you assign values to variables in R

VI. What is the purpose of the library() function in R?

VII. What are R scripts, and how are they executed?

VIII. How can you visualize data in R?

IX. How do you import and export data in R?

X. What is the role of the help() function in R?

XI. What is the significance of the NA value in R?

XII. What are the different ways to create a matrix in R?

XIII. What is the purpose of the summary() function in R?

XIV. How to create data frame in R.

XV. Write any 2 commands to plot the data 2,3,4,5,6,7

XVI. Why is sequence how to create it in r

XVII. Define array

XVIII. How to add comments in r

XIX. How to impot data in r

XX. What are some disadvantages of using R?

XXI. What are some disadvantages of using R?

XXII. How do you add a new column to a data frame in R?

XXIII. What is a factor in R?

XXIV. How to create a user-defined function in R?

XXV. What is the difference between the functions apply(), lapply(), and tapply()?

XXVI. Define the function of setwd().

Q1. Describe the operations for the following functions (10):

a) length()
b) sample()
c) matrix()
d) sd()
e) setwd()

Q2. Write R code to generate the following vectors or to perform the given task : (10)

a) Generate a sequence of 20 equally spaced numbers between 0 and 1.
b) Create a vector of logical values indicating whether each element of the sequence from (a) is greater than 0.5.
c) Find the sum of all elements in the vector from (a).
d) Create a new vector by multiplying each element of the vector from (a) by 10.
e) Find the maximum value in the vector created in (d).

Q3. Write an R code to perform the following tasks:
(10)

Height= 1,3,4,5,6,7,8

Weight=2,3,4,5,6,8,7

a) Create a scatter plot of Height vs. Weight.
b) Calculate the correlation coefficient between Height and Weight.
c) Fit a linear regression model predicting Weight based on Height.
d) Add the regression line to the scatter plot.
e) Extract the coefficients (intercept and slope) of the model.

Q4. Write an R code to generate the following vectors or to perform the given task: (10)

a) Generate 500 random observations from a Poisson distribution with lambda = 3.

b) Save the generated data as an .RData file.
c) Plot the barplot of the sample data.
d) Calculate the mean, variance, and skewness of the sample data and print the results.
e) Save the barplot as a .pdf file.

Q5 Given the following matrices

Write R code to perform the following tasks:

a) Calculate the matrix product of A and B. Report the resulting matrix.
b) Compute the transpose of matrix A.
c) Determine the determinant of matrix A.
d) Check if matrix A is invertible. If it is, calculate and report the inverse of matrix A.
e) Create a new matrix by binding matrix A and matrix B by columns.

Q6. Write R code to perform the following tasks using loop functions: (10)

a) Use a for loop to print the squares of numbers from 1 to 10.
b) Use a while loop to find the first number greater than 100 that is divisible by 7.